

# Toby Rodel - CV & Publication list

 TobyRodel |  Toby Rodel |  tobyrodel.github.io |  trodel01@qub.ac.uk |  +447860205787

## EDUCATION

---

|                             |                                                           |               |
|-----------------------------|-----------------------------------------------------------|---------------|
| October 2023 - March 2027   | PhD (Astrophysics) at <b>Queen's University Belfast</b>   | (In progress) |
| October 2022 - October 2023 | MSc by Research (Physics) at <b>University of Warwick</b> | (Completed)   |
| October 2019 - June 2022    | BSc (Physics) at <b>University of Warwick</b>             | (2:1)         |

## RESEARCH EXPERIENCE

---

**PhD (Queen's University Belfast with Prof Christopher Watson)** 2023 - March 2027 (est)

- Characterisation of long-period transiting planets and eclipsing binary systems including NGTS-EB-7 (Rodel et al. [2025](#)) and TIC-65910228 / NGTS-38 (Rodel et al. [2026](#)).
- Used PAWS/iSpec and isochrones to characterise host stars and Allesfitter to fit orbital solutions to transit and radial velocity data.
- Submitted succesful ESO proposal for multiband transit observations with ULTRACAM for validation.
- Continuation of MSc work predicting survey yields, including upgrades to custom pipeline.

**MSc by research (University of Warwick with Dr Daniel Bayliss)** 2022 - 2023

- Developed the Transit Investigation and Recoverability Application (TlRa) and used it to predict exoplanet discovery yields from *TESS*-SPOC data (Rodel et al. [2024](#)).
- Participated in eyeballing and vetting of *TESS* monotransit candidates.

**BSc project (University of Warwick with Dr Matteo Brogi)** 2021-2022

- Developed code to detect and fit secondary eclipses of known hot Jupiters in *TESS* data.
- Attempted albedo and heat redistribution factor determination.

## SKILLS & EXPERTISE

---

Research expertise: exoplanets, transits, radial velocities, wide field photometric surveys, long-period, low mass eclipsing binaries, stellar characterisation, survey yield prediction, occurrence rates.

Instruments: *TESS*, NGTS, ULTRACAM, CORALIE, HARPS, FEROS, LCOGT

Software: TlRa (custom pipeline), [Allesfitter](#), [PAWS](#), [iSpec](#), [isochrones](#), [Specmatch-emp](#), [AstroAriadne](#)

Programming languages: Python, SQL, HTML, CSS,  $\text{\LaTeX}$ , markdown

Other: public speaking and engagement, poster design, mentoring, acronym invention

## SERVICE & LEADERSHIP

---

**ESO DPR reviewer** P114 & P115

**NGTS communications chair** 2023 - 2025

- First chair of NGTS communications group to manage consortium promotion and outreach.
- Managed consortium social media and maintained webpage.
- Procured customised pins and stickers for consortium members.

**QUB Equitea committee member** 2023 - present

- Planned seminars and discussion sessions on Equity, Diversity and Inclusion topics and their relations to Astronomy

### **QUB ARC seminar committee member**

2024-2025

- Arranged for external speakers to give seminars and hosted them.

## **TEACHING AND MENTORING**

---

### **Co-supervisor of summer intern**

June-September 2026

- Helped design cross-disciplinary project extracting SNe lightcurves from NGTS data.
- On recruitment committee to rank of 14 applicants.
- Supervisor duties focussed on working with NGTS data and liaising with consortium.

### **Postgraduate teaching assistant - Queen's University Belfast**

2023-2024

- Problem class tutor for modules MTH1021 and PHY1002.

### **Mentoring of research students**

2023 - Present

- Assisted and mentored QUB MSci students with software, proofreading of reports and presentation practice.
- Assisted and mentored external students through the NGTS early careers group.

## **OUTREACH**

---

### **Planetarium programmes**

2022 - Present

- Setup and take-down of inflatable dome as well as presentation of shows using Stellarium and Universe Sandbox at various festivals and school visits.
- Volunteered at Warwick, cofounded new programme in Belfast.

### **Festivals and events**

Various

- **QUB Astronomy day: Volunteer 2024 & 2025**, on organising committee for 2026.
- **Astronomy on Tap Belfast** Speaker and Organiser / MC.
- **Big Bang Science Festival Birmingham**: Volunteer at Warwick booth.

### **Public talks and Lectures**

Various

## **REFEREES**

---

|                         |                                                 |                                                                      |
|-------------------------|-------------------------------------------------|----------------------------------------------------------------------|
| Prof Christopher Watson | PhD supervisor - Queen's University Belfast     | <a href="mailto:c.a.watson@qub.ac.uk">c.a.watson@qub.ac.uk</a>       |
| Dr Daniel Bayliss       | MSc supervisor - University of Warwick          | <a href="mailto:d.bayliss@warwick.ac.uk">d.bayliss@warwick.ac.uk</a> |
| Dr Sarah Casewell       | Collaborator - University of Leicester          | <a href="mailto:slc25@leicester.ac.uk">slc25@leicester.ac.uk</a>     |
| Dr Ernst de Mooij       | PhD 2nd supervisor - Queen's University Belfast | <a href="mailto:e.demooij@qub.ac.uk">e.demooij@qub.ac.uk</a>         |
| Dr Matteo Brogi         | BSc supervisor - University of Turin            | <a href="mailto:matteo.brogi@unito.it">matteo.brogi@unito.it</a>     |

## FIRST AUTHORED PUBLICATIONS

---

- Rodel, Toby et al. (July 2026). “TIC-65910228 b/NGTS-38 b, a 180 day transiting warm super-Jupiter”. In: MNRAS 549.4, stag1061. DOI: [10.1093/mnras/stag1061](https://doi.org/10.1093/mnras/stag1061).
- Rodel, Toby et al. (Feb. 2025). “NGTS-EB-7, an eccentric, long-period, low-mass eclipsing binary”. In: MNRAS 537.1, pp. 35–55. DOI: [10.1093/mnras/stae2799](https://doi.org/10.1093/mnras/stae2799).
- Rodel, Toby et al. (Mar. 2024). “TIARA TESS 1: estimating exoplanet yields from Years 1 and 3 SPOC light curves”. In: MNRAS 529.1, pp. 715–731. DOI: [10.1093/mnras/stae474](https://doi.org/10.1093/mnras/stae474).

## CO-AUTHORED PUBLICATIONS

---

- Apergis, Ioannis et al. (Jan. 2026a). “High-precision photometry with a scientific CMOS camera: II on-sky testing of the Marana camera at the NGTS facility”. In: *RAS Techniques and Instruments* 5, rzag022. DOI: [10.1093/rasti/rzag022](https://doi.org/10.1093/rasti/rzag022).
- Apergis, Ioannis et al. (Aug. 2026b). “NGTS-39 b: a 58 d transiting warm Jupiter in an eccentric orbit”. In: MNRAS 550.2, stag1271. DOI: [10.1093/mnras/stag1271](https://doi.org/10.1093/mnras/stag1271).
- Hobbs, Katlyn L. et al. (Aug. 2026). “The Fe I 4377 Å Line as a Solar Faculae Indicator: Insights from Spectral Ratio Analysis”. In: MNRAS. DOI: [10.1093/mnras/stag1462](https://doi.org/10.1093/mnras/stag1462).
- Kendall, Alicia et al. (Apr. 2026). “A 43 d transiting Neptune and two 25 d Saturns from TESS, NGTS, and ASTEP”. In: MNRAS 547.2, staf2189. DOI: [10.1093/mnras/staf2189](https://doi.org/10.1093/mnras/staf2189).
- O’Brien, Sean M. et al. (Dec. 2025). “NGTS-EB-8: A Double-lined Eclipsing M+M Binary Discovered by Citizen Scientists”. In: AJ 170.6, p. 316. DOI: [10.3847/1538-3881/ae0e0d](https://doi.org/10.3847/1538-3881/ae0e0d).
- Ulmer-Moll, S. et al. (Nov. 2025). “Detection and characterisation of a 106-day transiting Jupiter: TOI-2449 b/NGTS-36 b”. In: A&A 703, A258. DOI: [10.1051/0004-6361/202555168](https://doi.org/10.1051/0004-6361/202555168).
- Doyle, Lauren et al. (Apr. 2024). “The TESS-SPOC FFI target sample explored with Gaia”. In: MNRAS 529.2, pp. 1802–1813. DOI: [10.1093/mnras/stae616](https://doi.org/10.1093/mnras/stae616).
- Eschen, Yoshi Nike Emilia et al. (Dec. 2024). “Viewing the PLATO LOPS2 field through the lenses of TESS”. In: MNRAS 535.2, pp. 1778–1795. DOI: [10.1093/mnras/stae2427](https://doi.org/10.1093/mnras/stae2427).
- Hawthorn, Faith et al. (Feb. 2024). “TESS duotransit candidates from the Southern Ecliptic Hemisphere”. In: MNRAS 528.2, pp. 1841–1862. DOI: [10.1093/mnras/stad3783](https://doi.org/10.1093/mnras/stad3783).

## OBSERVING PROGRAMMES

---

### As PI

“Probing puffy exoplanet atmospheres with ULTRACAM” ESO, 118.25EL, ULTRACAM on NTT (19h)

### As Co-I

“Characterising transiting warm and temperate Jupiters to shed light on their formation” HARPS-N ITP 2026B & 2027A, HARPS-N (15N), RISE (70h), HiperCam (11h)

# CONFERENCE PRESENTATIONS

---

## Talks

- “Finding hidden long-period planets with *TESS*”, *UK Exoplanet Meeting*, University of Bristol, April 2026
- “How good is *TESS* at detecting long-period exoplanets”, *From Transits to Trends: the Next Decade of Long-Period Exoplanets*, University of New Mexico, August 2025
- “Long period planets from *TESS* and NGTS”, *Irish National Astronomy Meeting*, University of Galway, August 2024
- “Putting a TIaRA on SPOC”, *TESS Science Conference 3*, Massachusetts Institute of Technology, August 2024

## Posters

- “Eclipsing Binaries from the Next Generation Transit Survey”, *National Astronomy Meeting*, University of Birmingham, July 2026
- “Finding hidden long-period planets with *TESS*”, *National Astronomy Meeting*, University of Birmingham, July 2026
- “Finding hidden long-period planets with *TESS*”, *Exoplanets 6*, Instituto de Astrofísica e Ciências do Espaço, July 2026
- “NGTS-EB-7: One of the longest period EBLM systems ever found”, *Binary stars in the space era*, University of Keele, July 2025
- “NGTS-EB-7: One of the longest period EBLM systems ever found”, *UK Exoplanet Meeting*, University of Leeds, April 2025
- “Putting a TIaRA on SPOC”, *TESS Science Conference 3*, Massachusetts Institute of Technology, August 2024
- “Putting a TIaRA on SPOC: Predicting long-period planet yields from *TESS*”, *UK Exoplanet Meeting*, University of Birmingham, April 2024
- “Predicting the yield of long-period Planets in *TESS*”, *UK Exoplanet Meeting*, University College London, August 2023